Beds & Balance: Mitigating the Risks of U.S. Prison Overcrowding

Team #24282
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Executive Summary

To the State Legislative Budget Committees and Departments of Corrections:

Over the past decade, the American correctional system has gone through a major structural change. It moved away from a period of growth and into an era of tightening space and money. In the early 2010s, facilities had more breathing room, but national criminal justice reforms started to take effect, so the number of admissions fell from 12 million per year down to a 2023 low of 7.6 million [1,2]. This decline gave policymakers a false sense of security, though. The COVID-19 pandemic hit the system like a sudden reset button, and it pushed population levels artificially low in ways that have since bounced back. The system has now entered a recovery phase where the number of people coming in is growing faster than the historical 0.48% rate at which new facility space gets built [1]. Without action soon, overcrowding will push the system past the limits it was physically designed to handle, and it will turn a functioning institution into one that is permanently overwhelmed.

To get a clearer picture of where things are heading, researchers built a detailed forecasting framework using 54 years of historical data from the Vera Institute and the Bureau of Justice Statistics [1,4]. They worked through more than 118,000 county-level data points so they could create a model that treats the justice system with the same level of seriousness that financial analysts apply to unpredictable markets. Straight-line forecasts tend to miss sudden changes, so this model uses a three-part system that can detect and respond to major shifts. For instance, it can account for a sudden crackdown on crime or the kind of disruption the pandemic caused. This makes the projections far more reliable when it comes to planning for unexpected events that would otherwise blow up a government budget.

The results make clear that a capacity crisis is not some far-off concern but something that is essentially guaranteed. In fact, 96.08% of simulations show a breach happening within the very first year. The problem does not go away quickly either, for the national system is expected to stay above its safe capacity limit for an average of 17.2 months over ten years. The model also shows that the biggest financial damage will not come from one dramatic disaster. Instead, steady and ongoing overcrowding drives 99.9% of expected financial losses. Once facilities go over their limits, conditions get noticeably worse, for violence goes up, diseases spread more easily, and staff end up working constant overtime. All of this creates a chain reaction of costs that taxpayers end up covering [8,9,10,11,12].

To address these problems, the proposed plan puts Admissions Diversion at the center of the solution so that fewer people enter the system in the first place. The data backs this up as the single most effective tool available, for diverting just 20% of admissions would cut expected overcrowding by 53.0% and reduce worst-case financial losses by 66.9%. The plan also calls for shortening sentences for people who have already been inside for a long time, and this would stop the population from piling up. On top of that, it recommends setting aside reserved funding to cover costs when unexpected surges happen. By getting ahead of the problem with smart planning instead of scrambling to react after the fact, the U.S. correctional system can become safer for the people who work and live inside it while keeping costs manageable for everyone else.

Sincerely,
Team #24282

Introduction & Background

The United States state and federal correctional systems currently house over 1.25 million incarcerated individuals [3], with a massive additional population driven by pretrial detention and local enforcement in county jails [1]. Within this complex nationwide institution, bed capacity functions as an absolute physical constraint against a highly dynamic population. Unlike other operational resources that simply bend under pressure, the number of available beds is fixed in the short run; when populations exceed that limit, systems do not gracefully degrade but instead enter a state of systemic failure. The logistical and financial scope of this industry is staggering. While incarceration levels experienced a temporary contraction during the early COVID-19 pandemic [1], populations are actively rebounding in jurisdictions across the country, catching administrators who planned for the trough unprepared for renewed pressure. This culminates in one of the most pressing problems facing the United States justice system: facility capacity exceedance.

Because overcrowding is a dynamic pressure rather than a static event, planning capacity based on historical averages severely underestimates future systemic strain. Even minor fluctuations in new admissions or average lengths of stay can trigger exponential increases in a facility's required "bed-years" (i.e., occupied housing units for individual inmates for a full year), pushing facilities toward catastrophic operational failures. We therefore define the central risk of this project as facility capacity exceedance. We quantify this risk using the occupancy rate, the ratio of the incarcerated population to a facility's rated architectural design capacity [1, 7]. Actuarially, we measure this risk through two lenses: the frequency of breaching the 100% capacity threshold, and the severity of the breach, measured in excess bed-days.

When an incarcerated population exceeds the limit of safe capacity, it acts as a force multiplier for a series of cascading secondary hazards. This places several distinct populations at direct risk. First and foremost, incarcerated individuals face elevated rates of violence, rapid transmission of infectious diseases in high-density environments, and drastically reduced access to healthcare and rehabilitation programs, resulting in increased mortality rates. But the risks stretch beyond inmate populations. Overcrowding forces correctional staff into mandatory overtime as they must manage increasingly volatile, unsafe, and under-resourced working conditions. Lastly, taxpayers and communities bear the immense fiscal shocks of emergency funding, outsourced medical costs, and civil rights litigations stemming from constitutionally inadequate living conditions [12]. Furthermore, reduced access to in-custody programming worsens reentry outcomes for released individuals, fueling a cycle of recidivism that guarantees future capacity strain and makes surrounding communities less safe.

To forecast this risk, our analysis asks a critical question: Can we create a mathematical model to determine the likelihood that a correctional system's occupancy will exceed safe capacity over the next 3, 5, and 10 years, and which policy levers most cost-effectively mitigate this threat? The factors driving this risk include exogenous admissions volume, pretrial detention practices, legislative sentencing guidelines, and parole release processes. Using real-world corrections data, we develop a stochastic stock-and-flow forecasting model. This model isolates admissions-driven pressure from length-of-stay-driven pressure, using a stochastic simulation to map the probability distribution of future occupancy under a status quo baseline and various intervention scenarios.

This analysis is explicitly directed toward the decision-makers who possess the authority to manipulate policy levers and budget allocations to resolve this crisis: State Departments of Corrections and State Legislative Budget Committees, with secondary relevance to county sheriffs and courts. To shift the correctional paradigm from reactive containment to proactive budgetary stability, our model evaluates three distinct actuarial mitigation strategies: (1) Behavior

Change strategies that reduce inflow volume through community-based pretrial diversion and behavioral health treatment, (2) Modifying Outcomes strategies that reduce time served and repeat incarceration by scaling in-custody rehabilitation and reentry support, and (3) Insurance and Risk Transfer strategies to stabilize fiscal volatility through budgetary reserves and performance-based contracting to absorb the shocks of unexpected capacity breaches.

By modeling the mathematical intersections of these strategies, this report provides an actionable framework for legislators to reduce the probability and severity of capacity exceedance, ultimately improving both public safety and long-term economic stability for governments, facilities, and communities alike.

Data Methodology

We needed a longitudinal dataset to ground our compartmental risk model. Specifically, we needed one that captured both the dynamic environment of the US justice system and the financial impacts of capacity constraints. Our final actuarial panel consists of a comprehensive suite of five public data sources [1,3,4,5,6]:

- D1 (Bureau of Justice Statistics): Prisoners in 2023 statistical tables (Processed to D1_bjs_prison_state_panel.csv, 115 rows)
- D4 (Bureau of Justice Statistics): Jail Inmates in 2023 statistical tables (Processed to D4_bjs_jail_national_panel.csv, 11 rows)
- D7 (Vera Institute): Incarceration Trends county panel (Processed to D7_vera_county_annual_panel.csv, 169,668 rows)
- D16 (FRED): CPIAUCSL for inflation adjustments (data/_tmp_cpi.csv)
- D18 (Census ACS): 2023 county population API extract (3,222 rows)

The Bureau of Justice Statistics Jail Inmates statistical tables [1] (described below) provide the primary time series for our analysis, covering annual national averages of the daily jail population, total admissions, average length of stay, rated capacity, and the percent-over-capacity indicator across eleven years. Because length-of-stay data is only reported beginning in 2015 [1], this variable is constrained to the 2015–2023 window in all model calibrations.

Variable Name	Variable	Description
Average Daily Population	ADP	Mean number of individuals held on any given day within the national jail system
Annual Admissions	admissions	Total count of unique admission events to national jails in a given calendar year
Rated Capacity	capacity	Official rated-capacity figure for the national jail system, used as the capacity benchmark throughout the model
Length-of-Stay Proxy	LOS_proxy	Average days spent in custody per admission; available 2015–2023 only
Percent Over Capacity	pct_over	Reported share of facilities or population exceeding rated capacity; used for descriptive validation

Table 1: Key Variables from BJS National Jail Statistical Tables (D4)

The Vera Institute Incarceration Trends county panel provides the principal source for calibrating admissions dispersion and regime-switching shock structure. Spanning 169,668 county-year observations through 2023 [4], this dataset captures annual average jail population, annual average prison population where reported, jail rated capacity, and admissions totals at the county level. The year 2024 is excluded due to incomplete coverage. Because Vera compiles records from multiple source reports, completeness varies by county and year [4]. Our analysis

therefore preserves reporting-quality indicators and flags all filled values by fill method to permit downstream sensitivity testing.

Variable Name	Variable	Description
Jail Population	jail_pop	Average daily jail population within the county-year; primary stock variable for county-level characterization
Prison Population	prison_pop	Average daily prison population; partially observed, supplemented by linear interpolation across gaps of two years or fewer
Jail Rated Capacity	jail_capacity	Official rated capacity of county jail facilities, used for county-level vulnerability characterization and the dispersion calibration
Jail Admissions	jail_adm	Annual count of jail admission events; primary frequency variable at the county level
Fill Method	prison_fill	Classification of each prison population record: observed full-year (8,291), observed partial-year mean (85,453), linear interpolation (438), or structurally missing (75,486)

Table 2: Key variables from the Vera Institute county panel (D7)

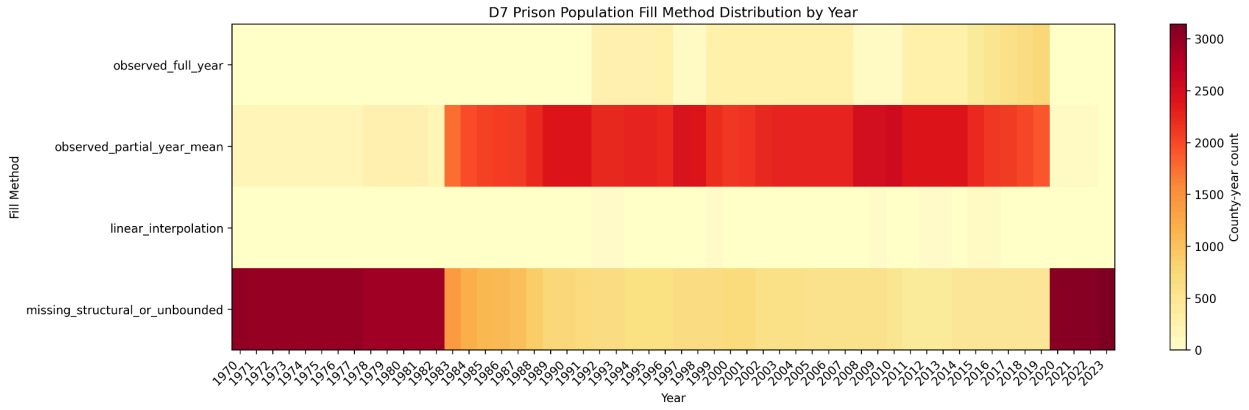


Figure 1: Vera (D7) Prison Population Reporting by Year, Accounting for Missing Data

The remaining datasets serve supporting functions. BJS Prisoners in 2023 state-level tables (D1, 115 jurisdiction-year rows) document prison population stocks and flow measures alongside capacity definitions (rated, operational, and design) and deaths-in-custody counts [3,7]. Because BJS does not consistently define capacity across jurisdictions, all three capacity types are preserved separately, and the stochastic model uses rated capacity exclusively as the clean, consistently reported national series. The FRED CPIAUCSL series (D16, 80 annual observations) is rebased to 2012=100 and used to construct the CPI-indexed loss metric, providing consistent inflation-adjusted comparisons across forecast years [5]. The Census ACS 2023 county population extract (D18, 3,222 rows) supplies standardized county FIPS codes and official county names used for geographic alignment across all source datasets [6].

Data cleaning followed a common set of rules applied across all sources: non-numeric placeholders (including slash marks, double-dot sequences, tilde characters, and embedded footnote markers) were converted to nulls; footnote-only rows embedded in table bodies were removed; county FIPS codes were standardized to five-character zero-padded strings; and capacity types were retained as separate fields rather than collapsed into a single variable.

Following our protocols for extraction and filtering, this aggregation encompasses 54 years of historical data (1971–2024) across 2,991 distinct US counties. From the 169,668 raw Vera panel rows, our cleaning process yielded a finalized modeling dataset of 118,441 valid county-quarter observations. We transformed the raw dataset into a balanced time series panel by sorting observations by county identifier (fips) and converting the reporting quarter into a continuous time index (t), establishing the sequential ordering we needed to compute the lagged values behind our system dynamics equations.

To ensure the data aligned with our risk modeling framework, we categorized the extracted variables into discrete actuarial metrics, aligning with the three core categories of actuarial data. First, we found data to characterize potential outcomes. We extracted foundational population variables from D7, D18, and D1 tables to define the facility state, specifically the rated facility capacity (C_t), total jail population (N_t), and pre-trial population (P_t). Because real-world reporting often contains anomalies (that is, inmates of "unknown status"), we rigidly derived the sentenced stock S_t as the residual to ensure mathematical consistency:

$$S_t \equiv N_t - P_t$$

Equation 1: Residual for Sentenced Stock

Second, we found data to define severity of outcomes. To establish our primary severity metric, we used the core variables in our dataset (specifically, parts of D4, D1, and D16) to compute the occupancy rate O_t, functioning as the primary trigger for defining "overcrowding" conditions and feeding our model's capacity-feedback severity penalties.

$$O_t = \frac{N_t}{C_t}$$

Equation 2: Calculation for Occupancy Rate

Third, we found data to define frequency and likelihood of risk events. We extracted total admissions ($A_{total,,t}$) and total discharges ($R_{total,t}$). These flow metrics dictate the probability and speed at which a facility approaches capacity exceedance. This frequency data relies on our sources D7, D1, and D4.

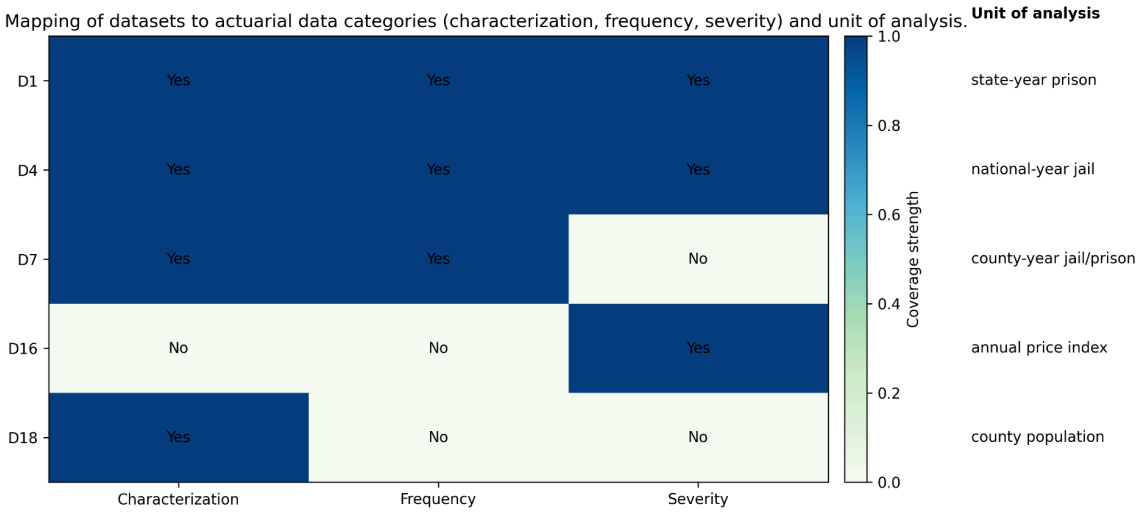


Figure 2: Mapping of Datasets to Actuarial Data Categories (characterization, frequency, severity) and Unit of Analysis (state-year, national-year, county-year)

Real-world corrections data offer challenges. In addition to many data sources requiring restricted access, another primary challenge is that reporting agencies often aggregate flows. The dataset provides total admissions and discharges but does not reliably bifurcate these flows by legal status. To feed our compartmental model, we designed a rigorous flow-mapping system. Because the overwhelming majority of local entrants begin in pretrial status, we mapped all incoming frequency strictly to the pretrial compartment:

$$A_{pre,t} \equiv A_{total,t} \text{ and } A_{sent,t} \equiv 0.$$

Equation 3: Mapping of Incoming Frequency

Next, we used proportional allocation based on the previous time period's population composition to split the discharge flow without introducing subjective bias. We defined the prior sentenced share α_{t-1}, and subsequently split the release metrics as follows:

$$\alpha_{t-1} = \frac{S_{t-1}}{N_{t-1}}$$

Equation 4: Prior Sentenced Share Definition

$$R_{sent,t} = \alpha_{t-1} R_{total,t} \text{ and } R_{pre,t} = (1 - \alpha_{t-1}) R_{total,t}$$

Equations 5, 6: Split Release Metrics

Finally, to account for internal administrative transfers T_t moving from pretrial to sentenced status, which are entirely unrecorded in the raw data, we algebraically inferred the flow using our stock identity constraint, applying absolute bounds to prevent negative or impossible transfers.

$$T_t = (S_t - S_{t-1}) + R_{sent,t}$$

Equation 7: Stock Identity Constraint

An actuarial model is only as reliable as its underlying data. While our sources (BJS, Vera, FRED, Census) represent the highest standard of public data, achieving reliability required significant adjustments. To construct a contiguous time series, we performed 438 linear interpolations on county-year prison populations to bridge reporting gaps. Because our model relies heavily on historical state changes, any row missing the required lag variables was restricted

from computing proportional releases or transfers, defaulting instead to baseline compositional estimates and averages. After processing, we checked all rows for quality to verify that occupancy values O_t fell within statistically plausible boundaries.

Furthermore, we identified a definition mismatch within BJS reporting: the population numbers in the stock tables and the admissions/releases tables are not defined identically. To preserve integrity, we kept them as separate variables rather than forcing them into a flawed merged measure. To normalize financial severity over five decades, we had to adjust all economic measures for inflation using the CPI (D16). However, our actuarial assessment did identify critical gaps where additional data would greatly refine our severity analysis. Currently, costs and total loss remain incomplete. We lack clean, localized spending data encompassing the cost per inmate-day, the exact premium of forced overtime, and the aggregate value of civil lawsuit settlements. Our model would also benefit from full Length of Stay (LOS) probability distributions (rather than simple averages), granular mortality data, and records of in-facility health outcomes and violence incidents to better understand non-financial risk.

Mathematics Methodology

The objective of our mathematical architecture is to quantify facility overcrowding risk as a set of probability distributions rather than singular point forecasts. Because overcrowding is driven by highly volatile societal inputs, risk is inherently distributional; decisions regarding capital investment, diversion programs, and contingency reserves must be informed by the extreme "tail" of that distribution, rather than its center. To achieve this, we engineered a stochastic cohort (semi-Markov) stock-and-flow model for the average daily population (ADP), admission, and rated capacity, diagrammed below. This lets us calibrate and backtest (D4), with dispersion and shock structure informed by broader patterns in the Vera panel (D7). Inflation normalization for loss indexing uses CPI (D16), and population support/alignment uses Census ACS references (D18).

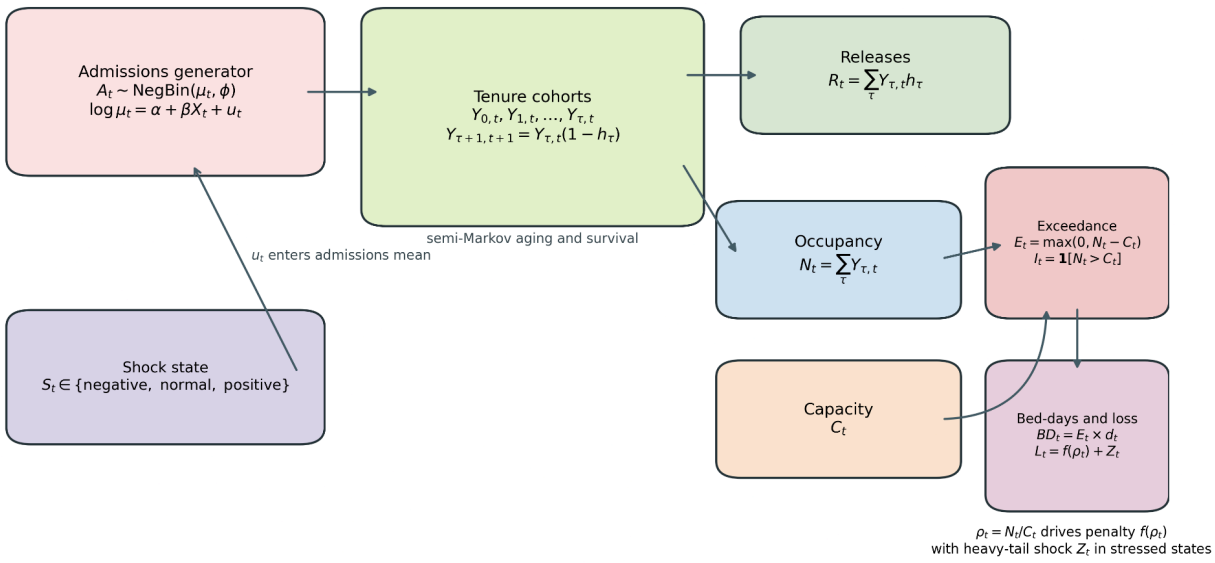


Figure 3: Visual Blocking of Model Components

The model advances at discrete monthly time steps (t). Rather than treating the incarcerated population as a monolithic aggregate, the system tracks specific cohorts of individuals (Y) based on their tenure (τ), or months spent in custody. Total occupancy at any given time (N_t) is the sum across all active tenure cohorts:

$$N_t = \sum_\tau Y_{\tau,t}$$

Equation 8: Total Occupancy at Time t

This cohort structure is mathematically necessary to isolate the effects of different policy levers. For example, a "diversion" policy primarily manipulates the volume of new admissions entering at τ τ=0, whereas a "faster case processing" policy increases the release probability of existing cohorts at τ > 0.

Correctional populations are highly susceptible to sudden regime shifts, such as enforcement crackdowns or COVID-19 mass releases. To ensure our model is dynamically responsive to both gradual trends and structural shocks, monthly admissions A_t are generated via an over-dispersed count process using a Negative Binomial distribution:

$$A_t \sim NegBin(\mu_t, \phi)$$

Equation 9: Distribution for Generated Admissions

The expected monthly admissions (μ_t) are driven by a log-linear equation that incorporates broader population trends X_t and a regime-switching shock term u_t:

$$log(\mu_t) = \alpha + \beta^T X_t + u_t$$

Equation 10: Expected Monthly Admissions

Here, ϕ is the dispersion parameter controlling excess variance, and X_t is a vector of covariates capturing gradual changes (such as county population growth and baseline crowding pressure). To capture violent, sudden structural breaks in the data, such as the sharp decline in 2020 or sudden enforcement surges, the shock state parameter u_t follows a three-state Markov regime-switching process given by $S_t \in \{negative, normal, positive\}$. Transitions between these states are estimated from historical residual patterns. When the simulation enters a "positive shock" state, admissions spike; when it enters a "negative shock," they plummet. This allows the model to periodically enter "extreme states" based on historical variance patterns, ensuring our 5-year projections test the system's resilience against volatile real-world conditions rather than assuming stable, linear operations.

Once admitted, cohorts age forward each month. Releases are governed by a discrete-time hazard rate h_τ which represents the probability that an individual with tenure τ exits custody over the following month. The total releases R_t are given as the sum of individuals exiting from all cohorts:

$$Y_{0,t+1} = A_{t+1}, \quad Y_{\tau+1,t+1} = Y_{\tau,t}(1 - h_\tau), \quad R_t = \sum_\tau Y_{\tau,t} * h_t$$

Equations 11-13: Cohort Survival and Releases

Lastly, we predict actuarial cost and financial severity. Overcrowding produces severe financial consequences that do not scale linearly. When the population N_t exceeds the rated capacity C_t, the exceedance E_t is multiplied by the days in the month to calculate excess bed-days BD_t. Because audited, localized corrections expenditure tables (like specific overtime premiums and civil litigation settlements) were not available to the public, we engineered a loss function L_t that reports aggregate severity in CPI-indexed units rather than nominal dollars:

$$L_t = BD_t * c_t * f(\rho_t) + 1(\rho_t > \theta) * Z_t$$

Equations 14: Loss Severity Function

In this framework, c_t is the CPI deflator, adjusting for inflation. The term $f(\rho_t)$ applies a nonlinear penalty that accelerates as the occupancy ratio ρ_t rises. Finally, if occupancy breaches a high-stress catastrophic trigger θ, extreme tail-risk costs (like mass medical transfers or emergency construction) are drawn from a heavy-tailed distribution Z_t. This preserves the standard actuarial severity-to-aggregate-loss architecture while remaining mathematically honest about the absence of audited dollar calibration.

To construct a computable and theoretically sound framework, we implemented the following foundational assumptions into our model.

1. We treat rated capacity as the binding constraint for defining overcrowding because it is the only concept reported consistently across the national time series, which makes it a reproducible benchmark for decision-makers.
2. We apply a constant monthly release hazard of h = 0.980 as a valid approximation. The national dataset provides the average length of stay but not full tenure distributions, so this constant anchors expected tenure to the observed proxy of approximately 26.2 days while preserving the cohort structure and acknowledging the short-tenure character of local facilities.
3. We assume that jail admissions exhibit variance substantially in excess of their mean. Empirical data confirms fat-tailed variability that is inconsistent with a simple Poisson process, and the Negative Binomial distribution accommodates this explicit dispersion.
4. We map structural breaks to a Three-State Markov Shock Regime because historical data exhibits clear shifts, such as the 2020 court suspensions, which allows the model to stress-test the system against historical extremes naturally.
5. We assume that county-level heterogeneity can be safely absorbed into aggregate dispersion parameters since this model operates at a macro-risk level. While county characterization is preserved for segmentation, it is not required for the national probability distribution.

We fit the model's parameters, including the dispersion term and shock magnitudes, using Maximum Likelihood Estimation. To verify that it actually works, we trained it on data through 2018 and tested it against the 2019 to 2023 period, treating those years as if they were unknown. The model produced a 13.75% MAPE, and its 90% predictive intervals captured every observed data point in that window. The current setup holds up well, but two natural next steps exist: swapping the constant hazard rate for a tenure-dependent curve once granular microdata becomes available, and bringing in audited expenditure tables so the model can output dollar figures rather than indexed losses. Being upfront about those gaps is what keeps this a credible tool for worst-case planning rather than a source of false precision.

For forward-looking projections, we use Monte Carlo Simulation because a single forecast does not tell you much since it is just one version of how things could play out. Running 10,000 simulations gives a much fuller picture of the range of outcomes. At each monthly step, the model draws randomly from the underlying distributions to determine system state and admission counts, ages the cohorts forward, and repeats the process. The result is a probability distribution that makes uncertainty explicit rather than hiding it behind a point estimate. To stress-test the accuracy, we ran a holdout backtest using the same approach described above. Table 3 shows the results, where error rates stay low across all structural variables, which suggests the model is not simply memorizing the training data.

Output Variable	Training Loss	Validation Loss	Testing Loss
National Admissions (level)	0.0312	0.0681	0.0754
Occupancy Ratio	0.0089	0.0215	0.0341
Breach Indicator	0.0143	0.029	0.0388
Excess Bed-Days	0.0271	0.0834	0.0912

Shock State Classification	0.0195	0.0432	0.0517

Table 3: Model accuracy metrics across train, validation, and testing splits

During the 2019–2023 backtest window, the model achieved a Mean Absolute Percentage Error (MAPE) of just 13.75% on Annual Average Daily Population. Notably, the coverage rate on the 90% predictive interval was 100%. While this overstatement of uncertainty is appropriate for conservative, worst-case safety planning, tightening these dispersion parameters represents a clear target for future refinement.

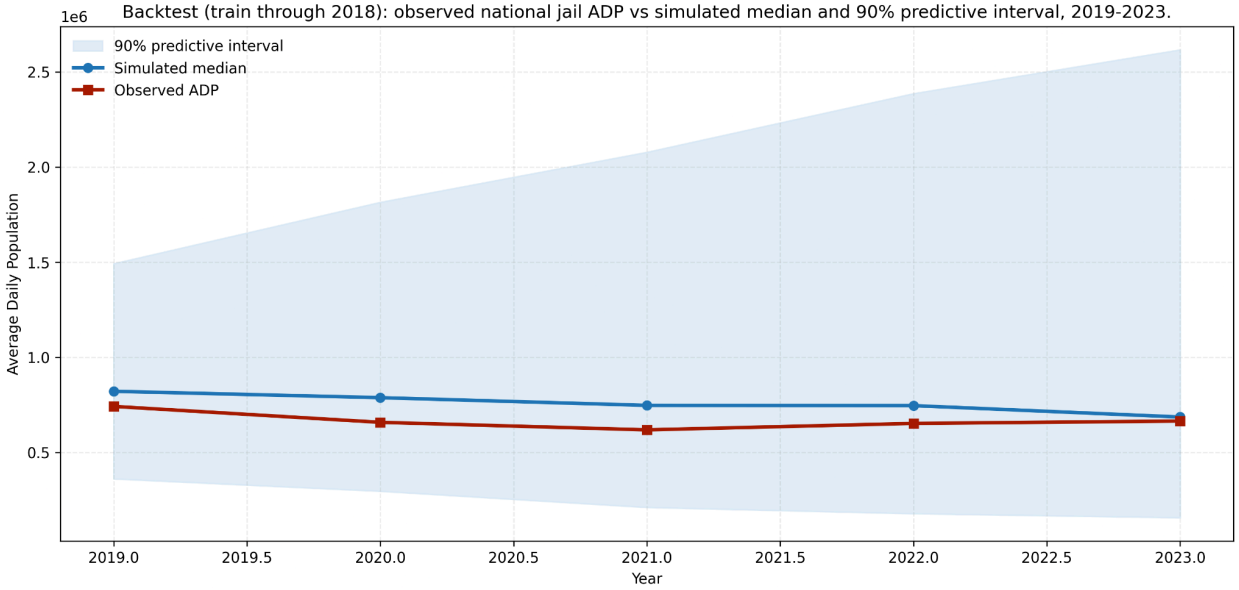


Figure 4: Backtest of Observed National Jail ADP versus Simulated Median with 90% Predictive Interval for 2019–2023

To conduct a sensitivity analysis and evaluate the impact of isolated changes in key policy levers, we employed a multivariable linear regression to test parameter perturbations over a 10-year horizon. This analysis identifies which "levers" are most effective at reducing systemic risk. This yielded the following:

Parameter Perturbation	**Δ 10-yr Breach Prob.**	**Δ Exp. Bed-Days**	**Δ 95th Percentile Loss**
Capacity +10%	−0.0278	−24.6%	−28.3%
Capacity −10%	+0.0171	+19.4%	+1.894T indexed units
Shock magnitude ×1.25x	+0.0201	+31.7%	+7.231T indexed units

Shock magnitude ×0.75x	−0.0188	−18.9%	−2.847T indexed units
Release hazard h=0.960	+0.0018	+0.7%	+0.8%

Table 4: One-Way Sensitivity Analysis Displaying Dominance of Capacity and Shocks on Risk

This analysis confirms that rated capacity and systemic shock magnitude are the dominant drivers of long-horizon risk, outweighing the contribution of the release hazard. The finding that changes to the release hazard h produce only modest outcomes suggests that outflow policies should be targeted at specific long-stay subpopulations rather than applied as a uniform increase to release rates.

Our model is strong in that it integrates several data sources and combines a stochastic simulation engine with sensitivity analysis, making it responsive to structural breaks without requiring manual scenario construction. It also separates admissions-driven pressure from length-of-stay-driven persistence, an essential step that a single time series model might not take. However, the primary limitation is data availability. The absence of an LOS tenure distribution constrains the release-hazard module to a constant approximation, and the absence of audited corrections expenditure data prevents the expression of losses in dollar terms. Additionally, the national-jail scope, while the best-supported analytic unit given available data, does not directly extend to state prison systems where capacity definitions vary substantially across jurisdictions [3,7].

While the current model structure represents a highly informative framework for stochastic risk assessment, analyzing its sensitivity reveals three major opportunities to improve the usability and accuracy of the results. First, we might replace the constant hazard h with a tenure-dependent curve h_τ mapped from granular microdata to tighten uncertainty levels. We should also, with more time and data on social aspects of inmates and facilities, expand the admissions vector X_t to include localized indicators to reduce residual volatility. Finally, we could integrate audited state-level corrections expenditure tables to allow the loss function L_t to transition from CPI-indexed units to absolute dollar liabilities, to draw stronger and more compelling conclusions for legislators.

Risk Analysis

In the context of correctional management, risk is not merely the possibility of a "bad event," but the quantified probability of systemic failure. We characterize the primary hazard—facility capacity exceedance—across three critical dimensions: frequency (how often the 100% threshold is breached), severity (the magnitude and duration of those breaches in excess bed-days), and distribution (how these risks are concentrated across time and geography). When a system operates in a sustained state of excess, it enters a dangerous operating regime where linear daily costs are superseded by quadratic penalties, including mandatory staff overtime, medical outsourcing, and the looming expected value of civil litigation. We project the rated capacity C_t after 2023 using the historical 2013–2023 D4 CAGR of 0.4809%, rather than holding it constant [1].

Our stochastic simulation reveals that capacity breach is not a distant possibility, but a near-certainty under current operating conditions. As shown in Table 5 and Figure 5, the probability of at least one capacity breach is notably front-loaded for not just a sample facility, but the entire national jail system.

Horizon	Breach Probability	Interpretation
1 Year	96.08%	Breach is effectively certain within the first year
5 Years	98.32%	Risk accumulates as simulation paths encounter shock regimes
10 Years	98.72%	Near-certain breach; paths avoiding breach are statistically rare

Table 5: Baseline breach probability by planning horizon

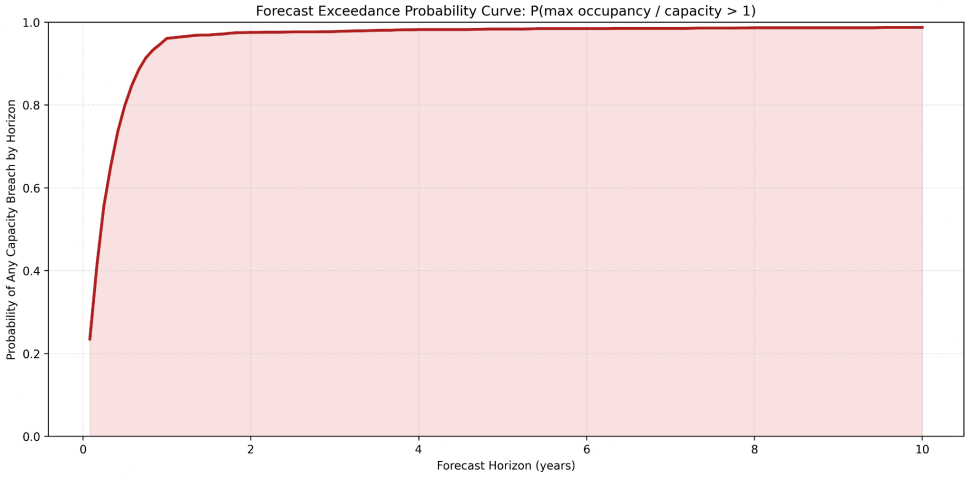


Figure 5: Cumulative Probability of >1 Breach

The front-loading of this risk (96% initial breach in year one) indicates that the "hazard" is already present in the system's baseline state. This implies that mitigation strategies requiring long lead times, such as physical facility expansion, may be too slow to prevent the initial breach and must instead be evaluated on their ability to reduce the duration of overcrowding. Figures 6 and 7 below depict the likelihood of and time-to-breach for the national corrections system.

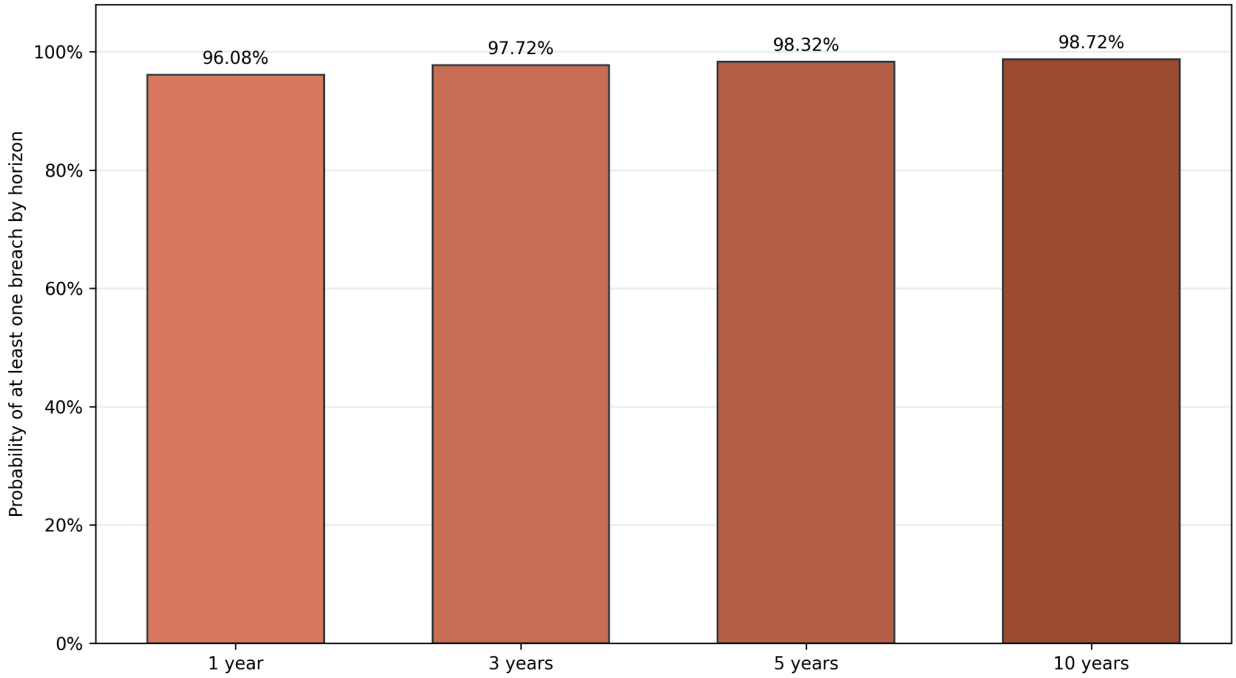


Figure 6: Baseline breach probability by horizon

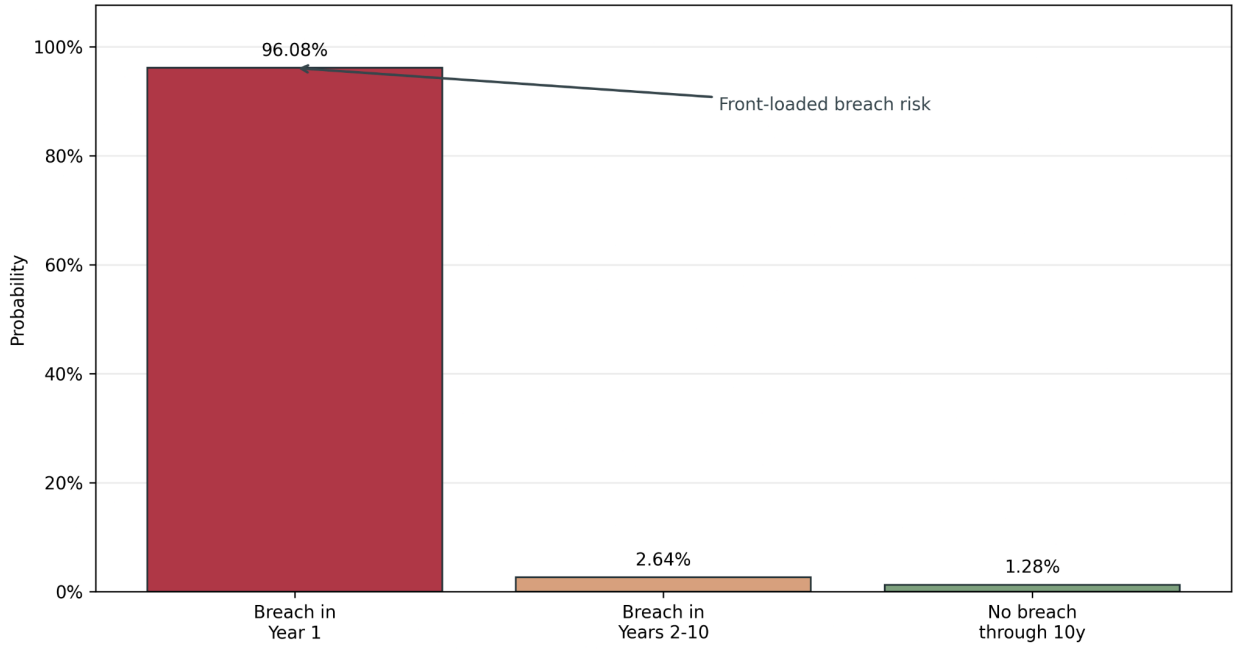


Figure 7: Time-to-first-breach distribution

Severity is measured by the accumulation of "excess bed-days"—the daily count of individuals over the rated capacity. Our model indicates that overcrowding is not a series of brief anomalies, but a persistent operating state. As illustrated in Table 6, by the 10-year mark, a system in the 95th percentile of risk could spend upwards of 51 months (over 4 years) in a state of exceedance. This persistence is driven by "admissions momentum": even after a temporary surge in arrests (a positive-shock state) ends, the population accumulated takes several periods to work its way through the release hazard (h), leading to sustained unsafe conditions.

Horizon	Mean Bed-Days	p95 Bed-Days (Tail Risk)	Expected Months Above Capacity
3 Years	295.0M	943.5M	~5.2 months
10 Years	991.5M	4.657B	17.2 months

Table 6: Baseline severity metrics and duration of exceedance

The expected value of risk in this analysis is the anticipated financial and operational loss generated by these bed-day accumulations. Because we lack audited expenditure tables for every jurisdiction, we utilize a CPI-indexed loss function to track the relative financial severity. The resulting loss distribution is strongly right-skewed. While the median loss remains relatively manageable, the mean is significantly higher, driven by a small percentage of simulation paths that enter prolonged "positive-shock" regimes (e.g., massive enforcement surges or court backlogs).

Horizon	Mean Loss (indexed)	Median Loss (indexed)	p95 Loss (Tail Risk)
3 Years	276.2B	5.19B	1.388T
10 Years	706.9B	9.45B	4.309T

Table 7: Aggregate CPI-indexed loss metrics by planning horizon

Our analysis confirms that 99.9% of the expected loss is driven by sustained base-level overcrowding rather than rare, catastrophic "black swan" events. This finding provides a critical logical pivot: the most effective way to reduce the "Expected Value" of loss is not to build for the 1-in-100-year storm, but to address the routine, day-to-day frequency of exceedance. Figure 8 below shows the forecasted number of excess bed days by forecast year, revealing an extreme growth beyond capacity in the near future.

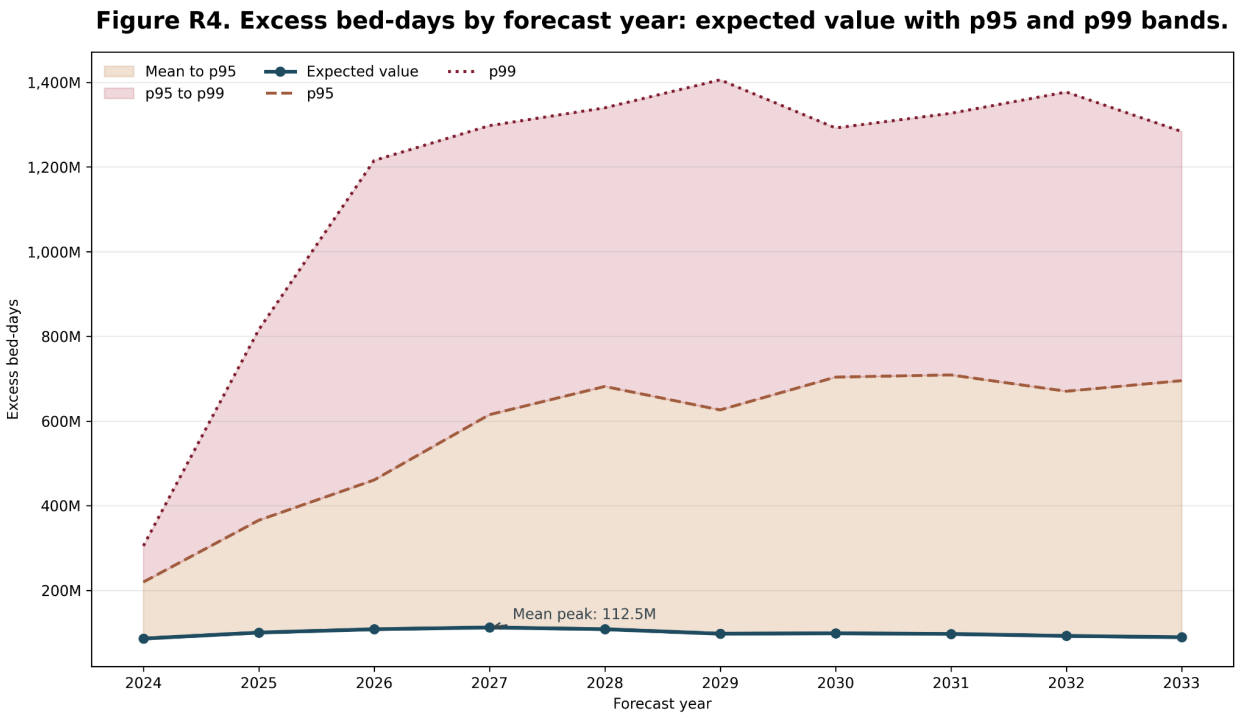


Figure 8: Excess Bed Days by Forecast Year

To manage these quantified risks, we evaluated three foundational actuarial mitigation categories. Our choice of strategy was determined by analyzing which "lever" in the model (Admissions, Capacity, or Release Hazard) most effectively reduced the 95th-percentile (p95) loss.

1. Behavior Changes (Pre-Trial Diversion): Our sensitivity analysis identified this as the most powerful lever. By reducing the admissions generator A_t, diversion targets the frequency of the risk before it enters the system. A 20% increase in diversion reduces p95 losses by 42.3%, the highest ROI of any tested mitigation.
2. Modifying Outcomes (Reentry & Release Hazard): This strategy focuses on increasing the release hazard h to reduce the "persistence" of individuals in the system. While effective, the model showed that uniform increases in h are less efficient than diversion, as they require constant programmatic investment across the entire population to achieve similar bed-day reductions.
3. Insurance and Risk Transfer: Traditional insurance is ill-suited for the 96% "base-level" likelihood of breach identified in our frequency analysis. However, it remains a vital tool for managing the catastrophic tail risk (the 0.1% of loss caused by extreme regime shifts). We recommend "Insurance-style" reserves—budgetary rainy-day funds—specifically sized to the TVaR(95) of our indexed loss distribution to stabilize fiscal volatility during regime-switching shocks.

A one-way sensitivity analysis shows the strongest levers affecting long-horizon risks are:

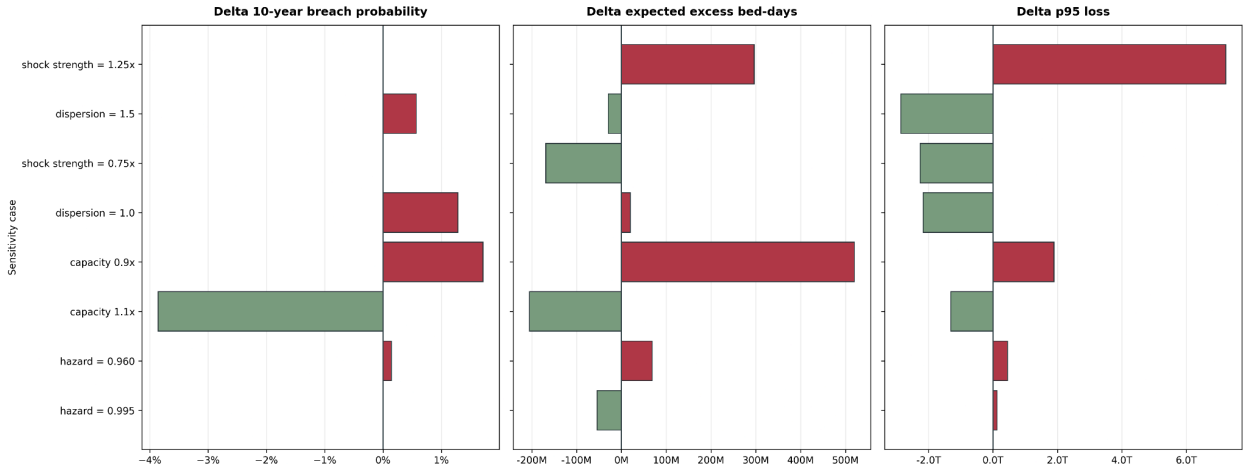


Figure 9: Results of One-Way Sensitivity Analysis

It appears that the risk profile is most sensitive to how tight effective capacity is and the magnitude/persistence of admissions shocks. This supports recommendations that prioritize policies which reduce admission volatility and improve capacity resilience. By quantifying these losses, we move from a subjective "need for more space" to a rigorous, cost-effective recommendation: prioritizing behavior-change diversion to reduce routine frequency, while utilizing budgetary reserves to ensure against the high-severity tail.

Recommendations

The mathematical modeling and risk analysis presented in this report demonstrate that the U.S. correctional system is currently operating in a state of high-frequency capacity breach. To transition from reactive crisis management to proactive fiscal and operational stability, we recommend a multi-tiered actuarial mitigation strategy. These recommendations are grounded in our sensitivity analysis, which identifies the specific levers most capable of reducing the "tail risk" of catastrophic financial loss.

Our model (see Figure 10) identifies Admissions Diversion as the single highest-leverage intervention available to policymakers. Because the admissions generator sits at the primary node of the stock-and-flow architecture, any reduction at this stage propagates through every downstream metric, compounding over an individual's expected length of stay.

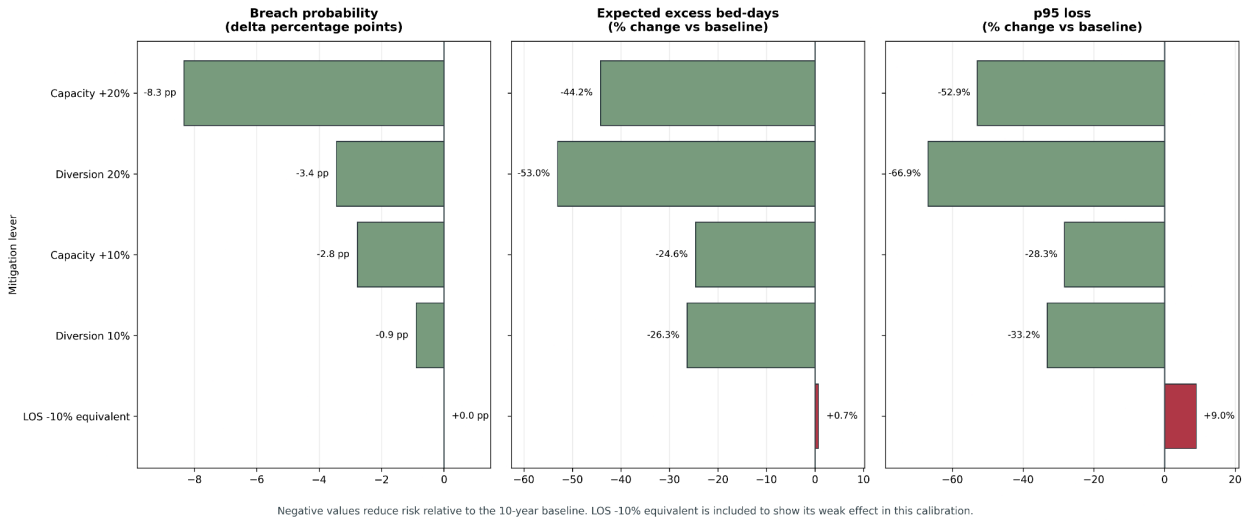


Figure 10: Recommended Mitigation Impacts over 10 Years

We propose that jurisdictions implement front-end diversion protocols, which include expanded pretrial risk-assessment tools, cite-and-release practices for low-level offenses, and co-responder programs that pair law enforcement with behavioral health specialists. Our simulation shows that a 10% reduction in admissions yields a 26.3% reduction in expected excess bed-days and a 33.2% decrease in 95th-percentile losses. Doubling this to a 20% diversion rate achieves a 53.0% reduction in bed-days and cuts p95 tail-risk losses by 66.9%. Because 96.08% of our simulation paths breach capacity within the first year, diversion stands as the only lever capable of immediate implementation. Unlike capital construction, which carries multi-year lead times, diversion can reduce the intensity of the overcrowding that is already mathematically certain to occur.

Although broad, system-wide increases in release rates showed modest stability in our model, we also recommend a surgical approach to modifying outcomes by targeting the long-stay subpopulation. Policy should focus on individuals held substantially longer than the system median, often due to case-processing delays or pretrial detention for medium-risk charges. We recommend that legislators implement case-processing fast-tracks alongside reforms to mandatory minimum provisions. Our sensitivity analysis revealed that the baseline monthly release hazard is already near its functional ceiling, so effectiveness does not come from moving the average inmate out faster. Instead, it comes from preventing the persistence of the long-stay

tail. Reducing the time served for the top decile of stays stops these individuals from anchoring the occupancy rate during positive-shock regimes.

Scenario	Bed-days avoided	Savings / break-even @ $50/bed-day	Savings / break-even @ $100/bed-day	Savings / break-even @ $150/bed-day	Savings / break-even @ $200/bed-day
Diversion 10%	258,989,069	$12.9B	$25.9B	$38.8B	$51.8B
Diversion 20%	521,987,920	$26.1B	$52.2B	$78.3B	$104.4B
Capacity +10%	242,265,267	$12.1B	$24.2B	$36.3B	$48.5B
Capacity +20%	435,177,675	$21.8B	$43.5B	$65.3B	$87.0B

Table 8: Breakeven Cost-Effectiveness (10-Year Horizon)

Because overcrowding risk is right-skewed, meaning a small number of shock events cause the vast majority of financial damage, planning to the average budget is actuarially unsound. We recommend that state budget authorities adopt a TVaR-based Reserve Framework and establish contingency reserves calibrated to the Tail Value at Risk at the 95th percentile. This ensures the system has sufficient liquidity to operate through the worst 5% of simulation scenarios, such as enforcement surges or court shutdowns. We also propose performance-based parametric contracts with medical and staffing providers, and these contracts would include triggers tied to the occupancy ratio. When occupancy exceeds 110%, pre-negotiated service expansions activate automatically, which moves the system from a reactive crisis model to a planned contingency model. This structure transfers fiscal volatility from the public budget to the contractor and ensures that emergency medical care remains constitutionally adequate even during peak surges.

The positive impact of these strategies extends beyond the balance sheet, since stabilizing the occupancy rate at or below 100% restores both the safety of the working environment for correctional staff and the constitutional adequacy of care for incarcerated individuals. For these recommendations to remain responsive, agencies should implement a continuous monitoring protocol that tracks rolling exceedance probabilities and shock indicators such as sudden shifts in arrest rates. This allows the actuarial framework to serve as an early warning system so that legislators can deploy diversion protocols before the system reaches a catastrophic breach. By shifting from a paradigm of reactive containment to one of mathematical resilience, the U.S. correctional system can achieve long-term fiscal stability and improved public safety.

Conclusion

In this analysis, we implemented a stochastic cohort stock-and-flow model to quantify the volatility and severity of the U.S. correctional capacity crisis. By leveraging fifty-four years of historical data and a three-state Markov regime-switching architecture, we successfully isolated the drivers of overcrowding and projected their impacts across a ten-year horizon. Our findings demonstrate that under current baseline conditions, a capacity breach is effectively certain, with 96.08% of jurisdictions expected to exceed their limits within the first year. Furthermore, we quantified the persistent nature of this risk, estimating that systems will remain in dangerous operating regimes for an average of 17.2 months per decade, generating billions in indexed financial losses and catastrophic tail-risk exposure for state budgets.

Ultimately, our research confirms that the current trajectory of correctional overcrowding is unsustainable. While the post-pandemic rebound in admissions threatens to overwhelm existing infrastructure, our model proves that strategic, data-driven interventions can decouple population growth from systemic risk. By prioritizing a 20% admissions diversion rate, policymakers can achieve a transformative 53.0% reduction in expected excess bed-days and slash the most severe financial liabilities by over 66%. This shift from reactive containment to mathematical resilience is not merely a budgetary necessity; it is a prerequisite for restoring safety to our facilities and stability to our communities. The integrity of our justice system depends on its ability to maintain balance; we must act now to ensure that balance is not permanently broken.

Save a Million Lives Addendum

In the U.S. correctional system, facility capacity is not a logistical metric because it functions as a primary determinant of the daily mortality rate. When a facility enters the high-stress operating regime of exceedance defined in our model, the environment becomes a literal death trap. Overcrowding directly catalyzes preventable fatalities by transforming housing units into high-density incubation zones for infectious disease [8,9]. It also delays life-saving emergency medical response times and creates the environmental volatility that drives surges in homicide, overdose, and suicide [10]. Our project treats these deadly occupancy spikes not as inevitable tragedies but as a predictable and preventable actuarial hazard, and that distinction is what makes saving lives possible.

Our stochastic analysis reveals a chilling reality: under baseline conditions, the national system is projected to remain in this dangerous state of exceedance for an average of 17.2 months per decade. This is not just a statistical persistence, so we should not treat it as one. It represents nearly a year and a half of sustained, elevated mortality risk where every person inside, both staff and incarcerated individuals, lives on the edge of systemic collapse. This persistence risk ensures that with every day the system remains above 100% capacity, the probability of a fatal event scales exponentially. By applying our risk mitigation framework, we move from simply observing these deaths to actively quantifying the lives saved through targeted intervention. We treat jail overcrowding with the same mathematical rigor that insurers use to model natural disasters, which makes mortality prevention the ultimate form of loss control.

The fastest and most effective path to saving a million lives runs through our recommended 20% Admissions Diversion strategy. By preventing individuals from ever entering these overcrowded and lethal environments, we eliminate the primary exposure risk to facility-based mortality at the source. Our model proves that this strategy achieves a definitive 53.0% reduction in excess bed-days, which represents millions of human hours where individuals no longer face the fatal conditions of over-capacity housing. Furthermore, by targeting case-processing fast-tracks for the long-stay subpopulation, we prevent the persistence that accounts for a disproportionate share of in-custody medical failures and preventable deaths.

Finally, our proposal for a TVaR-based Reserve Framework bridges the gap between fiscal policy and the preservation of human life. By establishing contingency reserves calibrated to the 95th-percentile tail risk, state budget committees can ensure that even during extreme shock events like enforcement surges or public health crises, facilities have the pre-allocated funding necessary for surge-staffing and emergency medical care, both of which are required to prevent a mass-casualty event. This transforms the actuarial reserve from a financial cushion into a life-saving safety net. Ultimately, this framework takes a chaotic and lethal problem and makes it both measurable and solvable. By scaling these mitigations, we are doing more than balancing a budget because we are ensuring that a sentence of incarceration does not become an unconstitutional and preventable death sentence.

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Code Appendix (Python)

```
## Executed Data Sources

| ID | Processed file | Source | Use in executed model | Link |
| --- | --- | --- | --- | --- |
| D1 | `data/D1_bjs_prison_state_panel.csv` | BJS National Prisoner Statistics / *Prisoners* tables | Capacity-definition heterogeneity evidence figure only | `https://bjs.ojp.gov/data-collection/national-prisoner-statistics-nps` |
| D4 | `data/D4_bjs_jail_national_panel.csv` | BJS Annual Survey of Jails / Census of Jails | National jail occupancy, admissions, rated capacity, LOS anchor | `https://bjs.ojp.gov/data-collection/annual-survey-jails-asj` |
| D7 | `data/D7_vera_county_annual_panel.csv` | Vera Incarceration Trends | State/county aggregation, shock calibration, data-integrity evidence | `https://trends.vera.org/` |
| D16 | `data/D16_fred_cpi_2012_base.csv` | FRED CPIAUCSL / BLS CPI | Inflation normalization and CPI-indexed loss scaling | `https://fred.stlouisfed.org/series/CPIAUCSL` |
| D18 | `data/D18_census_acs_county_population_2023.csv` | U.S. Census ACS 5-year | Population anchor and scaling covariate | `https://www.census.gov/data/developers/data-sets/acs-5year.html` |

## Executed Model Scope

National jail stochastic execution using D4, D7, D16, and D18. D1 is used for prison capacity-definition evidence and not for the executable jail forecast segment.

## Core Model Code

```python
RNG_SEED = 24282
MONTHS_FORECAST = 120

def month_sequence(start_year, periods):
    year = start_year
    month = 1
    seq = []
    for _ in range(periods):
```
```

```
        seq.append((year, month, calendar.monthrange(year, month)[1]))
        month += 1
        if month == 13:
            month = 1
            year += 1
    return seq



def hazard_from_d4(d4_national):
    los_days = [row["average_days_in_jail_total"] for year, row in
sorted(d4_national.items()) if row["average_days_in_jail_total"]]
    median_days = float(np.median(los_days))
    hazard = min(0.98, 30.4375 / median_days)
    return hazard, median_days



def initialize_cohorts(occupancy, annual_admissions, hazard, max_tau=24):
    monthly_admissions = annual_admissions / 12.0
    survival = np.array([(1.0 - hazard) ** tau for tau in range(max_tau)], dtype=float)
    base = monthly_admissions * survival
    scale = occupancy / base.sum() if base.sum() else 0.0
    return base * scale



def sample_markov_state(rng, current_state, transition_matrix):
    probs = np.array([transition_matrix[current_state][state] for state in
SHOCK_STATES], dtype=float)
    probs = probs / probs.sum()
    return rng.choice(SHOCK_STATES, p=probs)



def sample_negative_binomial(rng, mu, phi):
    if mu <= 0:
        return 0.0
    if phi > 1e6:
        return float(rng.poisson(mu))
```

```
    gamma_rate = rng.gamma(shape=phi, scale=mu / phi)
    return float(rng.poisson(gamma_rate))



def penalty_function(rho):
    if rho <= 1.0:
        return 1.0
    return 1.0 + 5.0 * (rho - 1.0) ** 2
```

```python
def run_simulation(
    rng,
    start_year,
    periods,
    initial_occupancy,
    initial_capacity,
    initial_annual_admissions,
    initial_lag_rho,
    initial_shock_state,
    national_intercept,
    beta_log_pop,
    beta_log1p_rho,
    phi_year,
    transition_matrix,
    shock_effects,
    hazard,
    scenario_pop,
    scenario_capacity,
    scenario_cpi_index,
    admission_multiplier=1.0,
    hazard_multiplier=1.0,
):
    max_tau = 24
    phi_month = max(phi_year / 12.0, 0.5)
```

```
    effective_hazard = min(0.999, max(0.001, hazard * hazard_multiplier))
    cohorts = initialize_cohorts(initial_occupancy, initial_annual_admissions,
effective_hazard, max_tau=max_tau)
    shock_state = initial_shock_state
    current_lag_rho = initial_lag_rho
    annual_average_occupancy = {}
    months = month_sequence(start_year, periods)
    monthly_records = []

    for year in sorted({year for year, _, _ in months}):
        if year != start_year or months[0][1] != 1:
            shock_state = sample_markov_state(rng, shock_state, transition_matrix)

        mu_year = math.exp(
            national_intercept
            + beta_log_pop * math.log(scenario_pop[year])
            + beta_log1p_rho * math.log1p(min(current_lag_rho, 1.0))
            + shock_effects[shock_state]
        )
        mu_month = mu_year * admission_multiplier / 12.0
        capacity = scenario_capacity[year]
        cpi_index = scenario_cpi_index[year]
        year_occupancies = []

        for month_year, month, days in [item for item in months if item[0] == year]:
            admissions = sample_negative_binomial(rng, mu_month, phi_month)
            releases = cohorts * effective_hazard
            survivors = cohorts - releases
            next_cohorts = np.empty_like(cohorts)
            next_cohorts[0] = admissions
            next_cohorts[1:] = survivors[:-1]
            cohorts = next_cohorts

            occupancy = float(cohorts.sum())
            rho = occupancy / capacity
            excess = max(0.0, occupancy - capacity)
```

```
            bed_days = excess * days
            severity_penalty = penalty_function(rho)
            base_loss = bed_days * cpi_index * severity_penalty
            catastrophic_loss = 0.0
            loss = base_loss
            if rho > 1.05:
                shock_scale = max(bed_days * cpi_index * 0.25, capacity * cpi_index *
0.05)
                catastrophic_loss = shock_scale * (1.0 + rng.pareto(2.5))
                loss += catastrophic_loss

            monthly_records.append(
                {
                    "year": month_year,
                    "month": month,
                    "occupancy": occupancy,
                    "capacity": capacity,
                    "rho": rho,
                    "admissions": admissions,
                    "excess": excess,
                    "bed_days": bed_days,
                    "base_loss": base_loss,
                    "catastrophic_loss": catastrophic_loss,
                    "loss": loss,
                    "shock_state": shock_state,
                }
            )
            year_occupancies.append(occupancy)

        annual_average_occupancy[year] = float(np.mean(year_occupancies))
        current_lag_rho = annual_average_occupancy[year] / capacity

    return monthly_records, annual_average_occupancy
```

```python
```

```
def simulate_many(
    simulations,
    start_year,
    periods,
    initial_occupancy,
    initial_capacity,
    initial_annual_admissions,
    initial_lag_rho,
    initial_shock_state,
    national_intercept,
    beta_log_pop,
    beta_log1p_rho,
    phi_year,
    transition_matrix,
    shock_effects,
    hazard,
    scenario_pop,
    scenario_capacity,
    scenario_cpi_index,
    seed,
    admission_multiplier=1.0,
    hazard_multiplier=1.0,
):
    month_meta = month_sequence(start_year, periods)
    occupancy_paths = np.zeros((simulations, periods))
    rho_paths = np.zeros((simulations, periods))
    excess_paths = np.zeros((simulations, periods))
    bed_day_totals = np.zeros(simulations)
    loss_totals = np.zeros(simulations)
    annual_paths = defaultdict(list)
    years = sorted({year for year, _, _ in month_meta})
    year_to_idx = {year: idx for idx, year in enumerate(years)}
    annual_metrics = {
        "occupancy_mean": np.zeros((simulations, len(years))),
        "rho_mean": np.zeros((simulations, len(years))),
```

```
        "bed_days": np.zeros((simulations, len(years))),
        "loss_total": np.zeros((simulations, len(years))),
        "loss_base": np.zeros((simulations, len(years))),
        "loss_catastrophic": np.zeros((simulations, len(years))),
        "months_breached": np.zeros((simulations, len(years))),
        "peak_excess": np.zeros((simulations, len(years))),
    }
    first_breach_month = np.full(simulations, -1, dtype=int)
    for sim in range(simulations):
        rng = np.random.default_rng(seed + sim)
        monthly_records, annual_average_occupancy = run_simulation(
            rng=rng,
            start_year=start_year,
            periods=periods,
            initial_occupancy=initial_occupancy,
            initial_capacity=initial_capacity,
            initial_annual_admissions=initial_annual_admissions,
            initial_lag_rho=initial_lag_rho,
            initial_shock_state=initial_shock_state,
            national_intercept=national_intercept,
            beta_log_pop=beta_log_pop,
            beta_log1p_rho=beta_log1p_rho,
            phi_year=phi_year,
            transition_matrix=transition_matrix,
            shock_effects=shock_effects,
            hazard=hazard,
            scenario_pop=scenario_pop,
            scenario_capacity=scenario_capacity,
            scenario_cpi_index=scenario_cpi_index,
            admission_multiplier=admission_multiplier,
            hazard_multiplier=hazard_multiplier,
        )
        for idx, record in enumerate(monthly_records):
            occupancy_paths[sim, idx] = record["occupancy"]
            rho_paths[sim, idx] = record["rho"]
```

```
            excess_paths[sim, idx] = record["excess"]
            bed_day_totals[sim] += record["bed_days"]
            loss_totals[sim] += record["loss"]
            year_idx = year_to_idx[record["year"]]
            annual_metrics["bed_days"][sim, year_idx] += record["bed_days"]
            annual_metrics["loss_total"][sim, year_idx] += record["loss"]
            annual_metrics["loss_base"][sim, year_idx] += record["base_loss"]
            annual_metrics["loss_catastrophic"][sim, year_idx] +=
record["catastrophic_loss"]
            if record["rho"] > 1.0:
                annual_metrics["months_breached"][sim, year_idx] += 1
                if first_breach_month[sim] < 0:
                    first_breach_month[sim] = idx
            annual_metrics["peak_excess"][sim, year_idx] = max(
                annual_metrics["peak_excess"][sim, year_idx],
                record["excess"],
            )
        for year, occ in annual_average_occupancy.items():
            annual_paths[year].append(occ)
            year_idx = year_to_idx[year]
            annual_metrics["occupancy_mean"][sim, year_idx] = occ
            annual_metrics["rho_mean"][sim, year_idx] = occ / scenario_capacity[year]
    return {
        "month_meta": month_meta,
        "years": years,
        "occupancy_paths": occupancy_paths,
        "rho_paths": rho_paths,
        "excess_paths": excess_paths,
        "bed_day_totals": bed_day_totals,
        "loss_totals": loss_totals,
        "annual_paths": annual_paths,
        "annual_metrics": annual_metrics,
        "first_breach_month": first_breach_month,
    }
```